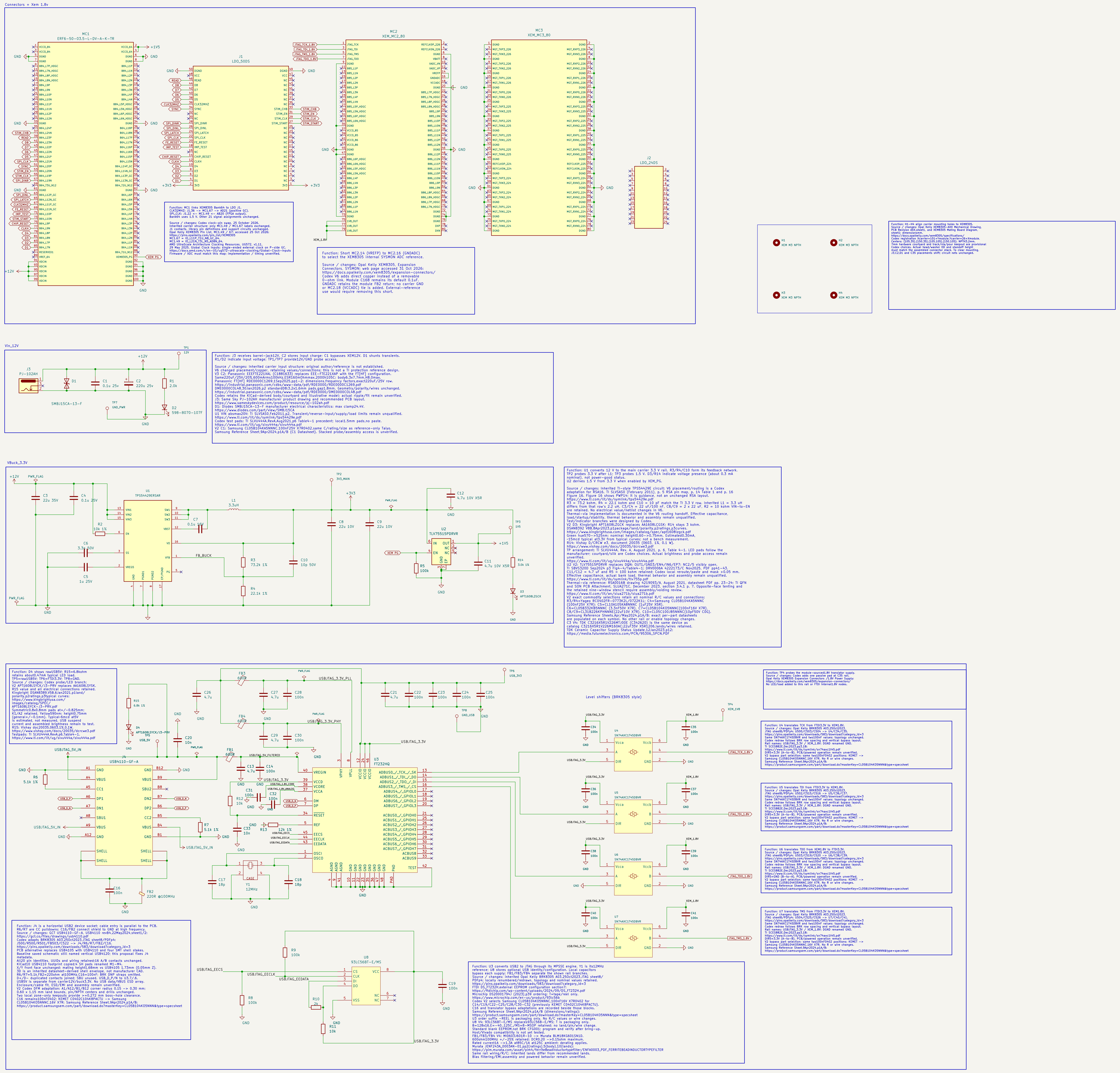

Function: MC1 links XEM8305 Bank64 to LDO J1.
CLK32MHZ: J1.36 -> MC1.67 -> AD21 (positive GC).
SPI_CLK: J1.22 <- MC1.49 <- AB20 (FPGA output).
Bank64 uses 1.5 V. Other 21 signal assignments unchanged.

Source / changes: Codex clock-pin swap, 25 October 2026.
Inherited carrier structure; only MC1.49 / MC1.67 labels exchanged.
J1 contacts, library pin definitions and support circuits unchanged.
Opal Kelly XEM8305 Pin List, MC1.49 / 67; accessed 25 Oct 2026:
https://pins.opalkelly.com/pin_list/XEM8305
MC1.67 = IO_L11P_T1U_N8_GC_64.
MC1.49 = IO_L21N_T3L_N5_AD8N_64.
AMD UltraScale Architecture Clocking Resources, UG572, v1.11,
29 May 2025, Global Clock Inputs: single-ended external clock on P-side GC.
https://docs.amd.com/r/en-US/ug572-ultrascale-clocking/Global-Clock-Inputs
Firmware / XDC must match this map; implementation / timing unverified.

Function: Short MC2.14 (VREFP) to MC2.16 (GNDADC)
to select the XEM8305 internal SYSMON ADC reference.

Source / changes: Opal Kelly XEM8305, Expansion
Connectors, SYSMON; web page accessed 31 Oct 2026:
https://docs.opalkelly.com/xem8305/expansion-connectors/
Codex V6 adds direct copper instead of a removable
0-ohm link. Module C168 remains its default 0.1uF.
GNDADC retains the module FB2 return; no carrier GND
or MC2.18 (VCCADC) tie is added. External-reference
use would require removing this short.

Function: H1-H4 align carrier mounting holes to XEM8305.
Source / changes: Opal Kelly XEM8305-AXX Mechanical Drawing,
PCB Revision AXX,sheet1, and XEM8305 Mating Board Diagram,
sheet1; dimensionsmm.
https://docs.opalkelly.com/xem8305/specifications/
Codex registration Xcarrier=101+Ymodule,Ycarrier=26+Xmodule.
Centers: (105,35),(150,35),(105,105),(150,105); NPTH3.2mm.
7mm hardware courtyard and track/via/pour keepout are provisional
Codex choices. Actual head/washer OD and standoff height
must match the assembled connector stack. To clear mounting,
J3,C2,D1 and C35 placements shift; circuit nets unchanged.

Function: J3 receives barrel-jack12V. C2 stores input charge; C1 bypasses XEM12V. D1 shunts transients.
R1/D2 indicate input voltage; TP1/TP7 provide12V/GND probe access.

Source / changes: Inherited carrier input structure; original author/reference is not established.
V6 changed placement/copper, retaining values/connections; this is not a TI protection reference design.
V3 C2: Panasonic EEEFTE221XAL (C1881633) replaces EEE-FTE221XAP with the FT[HF] configuration.
Same220uF/25V/20%,600mArms100kHz,ESR160mOhmmax,2000h105C; body6.3x7.7mm,H8.0max.
Panasonic FT[HF] RDE0000C1269,1Sep2025,pp1-2: dimensions,frequency factors,exact220uF/25V row.
https://industrial.panasonic.com/cdbs/www-data/pdf/RDE0000/RDE0000C1269.pdf
DME0000COL48,30Jan2026,p2 standardD8:3.2x1.6mm pads,gap1.8mm. Geometry/polarity/wires unchanged.
https://industrial.panasonic.com/cdbs/www-data/pdf/RDE0000/DME0000COL48.pdf
Codex retains the KiCad-derived body/courtyard and illustrative model; actual ripple/fit remain unverified.
J3: Same Sky PJ-102AH manufacturer product drawing and recommended PCB layout.
https://www.sameskydevices.com/product/resource/pj-102ah.pdf
D1: Diodes SMBJ15CA-13-F manufacturer electrical characteristics: max clamp24.4V.
https://www.diodes.com/part/view/SMBJ15CA
U1 VIN absmax20V: TI SLVSAS0,Feb2011,p2. Transient/reverse-input/supply/load limits remain unqualified.
https://www.ti.com/lit/ds/symlink/tps54429e.pdf
Codex test pads: TI SLVU444A,RevA,Aug2021,p6 Table4-1 precedent; local1.5mm pads,no paste.
https://www.ti.com/lit/ug/slvu444a/slvu444a.pdf
V2 C1: Samsung CL05B104KA5NNNC,100nF25V X7R0402,same C/rating/size as reference-only Taiyo.
Samsung Reference Sheet,9Apr2024,p1A/B (C1 Datasheet). Stacked probe/assembly access is unverified.

Function: U1 converts 12 V to the main carrier 3.3 V rail. R3/R4/C10 form its feedback network.
TP2 probes 3.3 V after L1; TP3 probes 1.5 V. D3/R14 indicate voltage presence (about 0.3 mA
nominal), not power-good status.
U2 derives 1.5 V from 3.3 V when enabled by XEM_PG.

Source / changes: Inherited TI-style TPS54429E circuit; V6 placement/routing is a Codex
adaptation for RSA16. TI SLVSAS0 (February 2011), p. 5 RSA pin map, p. 14 Table 1 and p. 16
Figure 16. Figure 16 shows PWP14; it is guidance, not an unchanged RSA layout.
https://www.ti.com/lit/ds/symlink/tps54429e.pdf
R3 = 73.2 kohm, R4 = 22.1 kohm and C10 = 10 pF match the TI 3.3 V row. Inherited L1 = 3.3 uH
differs from that row's 2.2 uH. C3/C4 = 22 uF/100 nF, C8/C9 = 2 x 22 uF, R2 = 10 kohm VIN-to-EN
are retained. No electrical value/netlist changes in V6.
Thermal-via implementation is documented in the V6 routing handoff. Effective capacitance,
load/startup/stability, thermal behavior and assembly remain unqualified.
Test/indicator branches were designed by Codex.
V2 D3: Kingbright APT1608LZGCK replaces AA1608LCGSK; R14 stays 3 kohm.
DSAN8392 V8B,8Apr2023,p1package/land/polarity,p2ratings,p3curves.
https://www.kingbrightusa.com/images/catalog/spec/apt1608lzgck.pdf
Green hue570->525nm; nominal height0.60->0.75mm. Estimated0.30mA,
~15mcd typical at3.3V from typical curves; not a bench measurement.
R14: Vishay D/CRCW e3, document 20035 (0603, 1%, 0.1 W).
https://www.vishay.com/docs/20035/dcrcwe3.pdf
TP arrangement: TI SLVU444A, Rev. A, August 2021, p. 6, Table 4-1. LED pads follow the
manufacturer; courtyard/silk are Codex choices. Actual brightness and probe access remain
unverified.
https://www.ti.com/lit/ug/slvu444a/slvu444a.pdf
U2 V2: TLV75515PDRVR replaces DQN; OUT1/GND3/EN4/IN6/EP7; NC2/5 visibly open.
TI SBVS320D Sep2024 p3 Fig4-4/Table4-1; DRV0006A 4222173/C Nov2025, PDF pp41-43.
C11/C12 = 4.7 uF and R5 = 100 kohm retained; Codex local reroute/paste and mask +0.05 mm.
Effective capacitance, actual bank load, thermal behavior and assembly remain unqualified.
https://www.ti.com/lit/ds/symlink/tlv755p.pdf
Thermal-via reference: RSA0016B drawing 4219093/A, August 2021, datasheet PDF pp. 23-24; TI QFN
and SON PCB Attachment, SLUA271C, December 2023, section 3.4.1 p. 7. Opposite-face tenting and
the retained nine-window stencil require assembly/voiding review.
https://www.ti.com/lit/an/slua271b/slua271b.pdf
V2 exact commodity selections retain all nominal R/C values and connections:
R3/R4=Yageo RC0402FR-0773K2L/0722K1L; C4=Samsung CL05B104KA5NNNC
(100nF25V X7R); C5=CL10A105KA8NNNC (1uF25V X5R),
C6=CL05B332KB5NNNC (3.3nF50V X7R), C7=CL05B104KO5NNNC(100nF16V X7R),
C8/C9=CL31B226KPHNNNE(22uF10V X7R), C10=CL05C100JB5NNNC(10pF50V C0G).
Samsung Reference Sheets,Apr/May2024,p1A/B; exact per-part datasheets
are populated on each symbol. No other rail or enable topology changes.
C3 V4: TDK C3216X5R1V226MTJ00E (C342620) is the same device as
catalog C3216X5R1V226M160AC;22uF35V X5R1206,lands/wires retained.
TDK Ceramic Capacitor Supply Status Update,12Jan2023,p12:
https://media.futureelectronics.com/PCN/95306_SPCN.PDF

Function: D4 shows rawUSB5V; R15=6.8kohm
retains about0.47mA typical LED load.
TP5=rawUSB5V; TP6=FTDI3.3V; TP8=GND.
Source / changes: Codex probe/LED branch;
V2 APT1608LSYCK/J3-PRV replaces AA1608LSYSK.
R15 value and all electrical connections retained.
Kingbright DSAN8389,V5B,6Jan2021,p1land/
polarity,p2ratings,p3typical curves:
https://www.kingbrightusa.com/
images/catalog/SPEC/
APT1608LSYCK-J3-PRV.pdf
Symmetric0.8x0.8mm pads at+/-0.825mm;
K1/A2 retained. Yellow590nm; height0.75mm
(general+/-0.1mm). Typical~6mcd at5V
is estimated, not measured. USB suspend
current and assembled brightness remain to test.
R15: Vishay doc20035,0603,1%,0.1W.
https://www.vishay.com/docs/20035/dcrcwe3.pdf
Testpads: TI SLVU444A,RevA,p6,Table4-1.
https://www.ti.com/lit/ug/slvu444a/slvu444a.pdf

Function: TP4 probes the module-sourced1.8V translator supply.
Source / changes: Codex adds one passive pad at C35 rail.
Opal Kelly XEM8305 Expansion Connectors /1.8V Power Supply:
https://docs.opalkelly.com/xem8305/expansion-connectors/
No LED/load added to this rail or FTDI internal1.8V nodes.

Function: U4 translates TCK from FTDI3.3V to XEM1.8V.
Source / changes: Opal Kelly BRK8305 A03,25Oct2023,
JTAG sheet8/PDFp4: U500/C503/C504 -> U4/C34/C35.
https://pins.opalkelly.com/downloads/583/download?category_id=3
Same SN74AXC1T45DBVR and two100nF values; topology unchanged.
Codex redraw follows BRK row spacing and vertical bypass layout.
Rail names: USBJTAG_3.3V / XEM_1.8V; DGND renamed GND.
TI SCES882E,Dec2023,pp3,18:
https://www.ti.com/lit/ds/symlink/sn74axc1t45.pdf
DIR5=3.3V (A-to-B). PCB/powered operation remain unverified.
V2 bypass part selection: same two100nF0402 positions; KEMET ->
Samsung CL05B104KO5NNNC,16V X7R. No R or wire changes.
Samsung Reference Sheet,9Apr2024,p1A/B:
https://product.samsungsem.com/part/download.do?masterKey=CL05B104KO5NNN&type=specsheet

Function: U5 translates TDI from FTDI3.3V to XEM1.8V.
Source / changes: Opal Kelly BRK8305 A03,25Oct2023,
JTAG sheet8/PDFp4: U502/C515/C516 -> U5/C36/C37.
https://pins.opalkelly.com/downloads/583/download?category_id=3
Same SN74AXC1T45DBVR and two100nF values; topology unchanged.
Codex redraw follows BRK row spacing and vertical bypass layout.
Rail names: USBJTAG_3.3V / XEM_1.8V; DGND renamed GND.
TI SCES882E,Dec2023,pp3,18:
https://www.ti.com/lit/ds/symlink/sn74axc1t45.pdf
DIR5=3.3V (A-to-B). PCB/powered operation remain unverified.
V2 bypass part selection: same two100nF0402 positions; KEMET ->
Samsung CL05B104KO5NNNC,16V X7R. No R or wire changes.
Samsung Reference Sheet,9Apr2024,p1A/B:
https://product.samsungsem.com/part/download.do?masterKey=CL05B104KO5NNN&type=specsheet

Function: U6 translates TDO from XEM1.8V to FTDI3.3V.
Source / changes: Opal Kelly BRK8305 A03,25Oct2023,
JTAG sheet8/PDFp4: U503/C519/C520 -> U6/C38/C39.
https://pins.opalkelly.com/downloads/583/download?category_id=3
Same SN74AXC1T45DBVR and two100nF values; topology unchanged.
Codex redraw follows BRK row spacing and vertical bypass layout.
Rail names: USBJTAG_3.3V / XEM_1.8V; DGND renamed GND.
TI SCES882E,Dec2023,pp3,18:
https://www.ti.com/lit/ds/symlink/sn74axc1t45.pdf
DIR5=GND (B-to-A). PCB/powered operation remain unverified.
V2 bypass part selection: same two100nF0402 positions; KEMET ->
Samsung CL05B104KO5NNNC,16V X7R. No R or wire changes.
Samsung Reference Sheet,9Apr2024,p1A/B:
https://product.samsungsem.com/part/download.do?masterKey=CL05B104KO5NNN&type=specsheet

Function: U7 translates TMS from FTDI3.3V to XEM1.8V.
Source / changes: Opal Kelly BRK8305 A03,25Oct2023,
JTAG sheet8/PDFp4: U504/C525/C526 -> U7/C40/C41.
https://pins.opalkelly.com/downloads/583/download?category_id=3
Same SN74AXC1T45DBVR and two100nF values; topology unchanged.
Codex redraw follows BRK row spacing and vertical bypass layout.
Rail names: USBJTAG_3.3V / XEM_1.8V; DGND renamed GND.
TI SCES882E,Dec2023,pp3,18:
https://www.ti.com/lit/ds/symlink/sn74axc1t45.pdf
DIR5=3.3V (A-to-B). PCB/powered operation remain unverified.
V2 bypass part selection: same two100nF0402 positions; KEMET ->
Samsung CL05B104KO5NNNC,16V X7R. No R or wire changes.
Samsung Reference Sheet,9Apr2024,p1A/B:
https://product.samsungsem.com/part/download.do?masterKey=CL05B104KO5NNN&type=specsheet

Function: J4 is a horizontal USB2 device socket; cable entry is parallel to the PCB.
R6/R7 are CC pulldowns; C16/FB2 connect shield to GND at high frequency.
Source / changes: GCT USB4110-GF-A, USB4110 revB4,22May2024,sheet1/2:
https://gct.co/files/drawings/usb4110.pdf
Codex adapts BRK8305 A03,25Oct2023,JTAG sheet8/PDFp4:
J500/R500/R501/FB503/C522 -> J4/R6/R7/FB2/C16.
https://pins.opalkelly.com/downloads/583/download?category_id=3
PCB alternative replaces USB4105 with USB4110 and four SMT shell stakes.
Baseline saved schematic still named vertical USB4120; this proposal fixes J4
metadata.
All20 pin identities, UUIDs and wiring retained;16 A/B contacts unchanged.
KiCad10 USB4110 footprint copied;4 SH pads renamed M1-M4.
X/Y front face unchanged; mating height1.68mm vs USB4105 1.73mm (0.05mm Z).
3D is an inherited datasheet-derived shell envelope, not manufacturer CAD.
R6/R7=5.1k,FB2=220ohm at100MHz,C16=100nF; BRK DNP straps omitted.
D+/D- duplicated contacts joined; SBU unused. USB_D_P/N to U3.7/.6.
USB5V is separate from carrier12V/buck3.3V. No USB data/VBUS ESD array.
Enclosure/cable fit, ESD/EMI and assembly remain unverified.
V2 Codex DFM adaptation: A1/A12/B1/B12 corner radius 0.15 -> 0.30 mm;
0.60 x 1.15 mm land bounds, pin/NPTH centers and drills unchanged.
Two local zone-only keepouts provide >=0.272 mm boss-hole clearance.
C16 remains100nF0402; KEMET C0402C104K8PACTU -> Samsung
CL05B104KO5NNNC,16V X7R; Samsung Reference Sheet,9Apr2024,p1A/B:
https://product.samsungsem.com/part/download.do?masterKey=CL05B104KO5NNN&type=specsheet

Function: U3 converts USB2 to JTAG through its MPSSE engine. Y1 is its12MHz
reference; U8 stores optional USB identity/configuration. Local capacitors
bypass each supply; FB1/FB3/FB4 separate the shown rail branches.
Source / changes: inherited Opal Kelly BRK8305 A03,25Oct2023,JTAG sheet8/
PDFp4; locally renumbered/redrawn, topology and nominal values retained.
https://pins.opalkelly.com/downloads/583/download?category_id=3
FTDI DS_FT232H,external EEPROM configuration section7:
https://ftdichip.com/wp-content/uploads/2024/09/DS_FT232H.pdf
Microchip DS20001794J (2023),p39 ordering: T=tape/reel only.
https://www.microchip.com/en-us/product/93lc56b
Codex V2 selects Samsung CL05B104KO5NNNC,100nF16V X7R0402 for
C14/C19/C22-C25/C28/C30-C32 (previously KEMET C0402C104K8PACTU).
C16 and translator bypass adaptations are recorded beside those blocks.
Samsung Reference Sheet,9Apr2024,p1A/B (dimensions/ratings):
https://product.samsungsem.com/part/download.do?masterKey=CL05B104KO5NNN&type=specsheet
U3 order suffix -REEL is packaging only. No R/C values or wire changes.
U8 V4: 93LC56BT-E/MS replaces93LC56B-E/MS; T is packaging only.
B=128x16,E=-40..125C,/MS=8-MSOP retained; no land/pin/wire change.
Standard blank EEPROM,not BRK CFG001; program and verify after bring-up.
Host/Vivado compatibility is not yet tested.
FB1/FB3/FB4 V4: MI0603J601R-10 -> Murata BLM18KG601SN1D.
600ohm100MHz +/-25% retained; DCR0.20 ->0.15ohm maximum.
Rated current1A ->1.3A at85C/1A at125C ambient; derating applies.
Murata JENF243A_0003AN-01,pp2(ratings),5(body),10(lands):
https://pim.murata.com/asset/pim4/ferriteBeadInductortypefilter/ENFA0003_PDF_FERRITEBEADINDUCTORTYPEFILTER
Same rail wiring/R/C; inherited lands differ from recommended lands.
Bias filtering/EMI,assembly and powered behavior remain unverified.

Sheet: /
File: Carrier_V1.kicad_sch

Title: XEM8305 / LDO Carrier

Size: A0
Date: 2026-10-31
KiCad E.D.A. 10.0.6
Rev: V6
Id: 1/1